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Q1) Attempt the following short answer question.

(Any four).

a) what is an IP address?

An IP address (Internet protocol address) is a unique numerical identities assigned to every devices on a network.

It allows devices to send and receive data and provides both location and identification of the devices in the network.

c) Define Subnet mask?

A Subnet mask is a 32 bit number that divides an IP address into a network portion and host portion.

It helps routers identify which part of the IP address refers to the network and which part identifies the devices within the network.

d) what is CIDR Notation? Give an example.

CIDR (classless inter-domain routing) notation represents an IP address along with its network prefix length.

It's written as:

IP Address / network bits (number)

Example:

192.168.1.0/24 (means first 24 bits are network bits).

h) what is the loopback ip address and why is it used?

- The loopback ip address is 127.0.0.1
- It is used to test the local machine network stack, check TCP/IP installation and ensure the device can communicate with itself without using the network interface.

Q2) for the following given ip's find class of ip, network ID and Host ID (any four).

classful details:

CLASS (A) : first octet (1-126) (/8)

Network = first octet.

Host = Last (3) octet.

CLASS (B) : first octet (128-191) (/16)

Network = first (2) octet.

Host = Last (2) octet.

CLASS (C) : first octet (192-223) (/24)

Network = first (3) octet.

Host = Last (1) octet.

a) 150.100.8.2

150.100.8.2

① ② ③ ④

① → 150 → (128-191) → B

Result:

Class of IP : (B)

Network ID : 150.100.0.0

Host ID : 0.0.8.2

b) 202.54.12.1
 200.54.12.1
 ① ② ③ ④

① → 200 → (192 - 223) → C.

Result:

Class of IP : (C).

Network ID : 202.54.12.0.

Host ID : 0.0.0.1.

c) 65.34.22.11

65.34.22.11
 ① ② ③ ④

① → 65 → (1 - 126) → A

Result:

Class of IP : (A).

Network ID : 65.0.0.0.

Host ID : 0.34.22.11.

d) 140.10.5.6

140.10.5.6
 ① ② ③ ④

① → 140 → (128 - 191) → B

Result:

Class of IP : (B).

Network ID : 140.10.0.0.

Host ID : 0.0.5.6.

- Q3) For the IP Address. (200.10.20.0/22), calculate
 B) number of IP's, network address, broadcast address.

Given:

IP: 200.10.20.0

prefix: 22.

1) Convert /22 $\rightarrow$ subnet mask.

(/22) means (22)(1) and (10)(0) in a 32 bit.

11111111.11111111.11111100.00000000.

↓ ↓ ↓ ↓
 255 255 252 0.

1 1 1 1 1 1 0 0
 (128) (64) (32) (16) (8) (4) (2) (1)

$$128 + 64 + 32 + 16 + 8 + 4 = 252 \text{ // partial. (3) octet.}$$

So Subnet mask is 255.255.252.0 //

2) Determine block size.

$$\begin{aligned} \text{block size} &= 256 - (\text{value of mask octet that is partial}) \\ &= 256 - 252 \\ &= 4. \end{aligned}$$

partial octet (3) $\Rightarrow$ 20.

Steps $\Rightarrow$ 4.

$$0 \Rightarrow 0 - 3$$

$$4 \Rightarrow 4 - 7$$

$$8 \Rightarrow 8 - 11$$

$$12 \Rightarrow 12 - 15$$

$$16 \Rightarrow 16 - 19$$

$$20 \Rightarrow 20 - 23 \leftarrow (20) //$$

$$24 \Rightarrow 24 - 27$$

3) Find network address.

2nd octet = 20. set all host bit to 0

$$20 \Rightarrow 20 - 23 \Rightarrow (20).$$

↓

20 is network address.

So 200.10.20.0 is already the network address.

4) Find broadcast address.

Broadcast is last address in the subnet.

set all host bit to 1. which mean 4th octet (255)

We know the block size = 4 (steps)

network address = 20

$$\text{broadcast address} = \text{network address} + (\text{steps} - 1)$$

$$= 20 + (4 - 1)$$

$$= 23 //$$

$$\text{broadcast address} = \underline{200.10.23.255}$$

5) Number of IP address.

$$\text{Host bit} = 32 - 22 \Rightarrow 10$$

$$= 2^{10} \Rightarrow 1024$$

Total number of IP = 1024.

$$\text{excluding (network \& broadcast)} = 1024 - 2$$

$$= 1022$$

6) Usable host IP's are.

$$\text{first : network} + 1 \Rightarrow 200.10.20.1$$

$$\text{last : broadcast} - 1 \Rightarrow 200.10.23.254 //$$

total IP : 1024

network address : 200.10.20.0

broadcast address : 200.10.23.255

Q4) If the given IP address is 192.168.1.77/26.

B) find the subnet range, first host, last host, broadcast.

Given:

IP: 192.168.1.77

prefix: 26.

1) subnet mask.

/26 $\rightarrow$ 26 (1) and 6 (0). $\Rightarrow 32 - 26 = 6$

11111111.11111111.11111111.11000000.

255

255

255

↓

1 1 0 0 0 0 0 0

(128) (64) (32) (16) (8) (4) (2) (1)

1 1

↓

128 + 64 $\rightarrow$ 192 (partial) (4) octet //

2) block size.

block size = 256 - (value of mask octet that is partial)

= 256 - 192

= 64 //

partial octet (4) IP = 77.

Steps = 64.

0 $\Rightarrow$ 0 - 63

64 $\Rightarrow$ 64 - 127 $\leftarrow$ (77). Start = 64; end = 127.

128 $\Rightarrow$ 128 - 191

3) network address.

192.168.1.64 //

4) Broadcast address.

Block size = 64.

Network address = 64.

$$\begin{aligned} \text{Broadcast address} &= \text{network address} + (\text{step} - 1) \\ &= 64 + (64 - 1) \\ &= 64 + 63 \\ &= 127 // \end{aligned}$$

broadcast IP = 198.162.1.127 //

5) Number of IP's.

$$\text{Host bit} = 32 - 26 = 6.$$

$$= 2^6 \Rightarrow 64 //$$

6) Usable IPs.

First usable : network + 1 = 198.162.1.65

Last usable : broadcast - 1 = 198.162.1.126.

Result:

Subnet Range : (198.162.1.64 → 198.162.1.127)

Network address : 198.162.1.64.

Broadcast address : 198.162.1.127.

First usable address : 198.162.1.65

Last usable address : 198.162.1.126.

total IP's : 64.

usable IP's : 62.

Q5). Describe the concept of Subnetting, and it's.

(A) advantages :

Subnetting :

- Subnetting is the process of combining multiple smaller networks (subnets) into a larger address block.

- It merges networks by reducing the number of network bits, and increasing the host portion.

- It was introduced to reduce the large number of routing table entries on the internet.

example: combine 4 class (C) networks:

192.168.4.0

192.168.5.0

192.168.6.0

192.168.7.0

These can be summarized as:

192.168.4.0/22

Advantages:

1. Reduces routing table size.
2. Improve router performance.
3. Faster routing decision.
4. Less bandwidth used for routing updates.
5. Simplifies network management.

Subnetting provides efficient, scalable and optimized routing, especially for large networks and ISPs.